# Jesús M. Rueda-Becerril, Ph.D.

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## Professional Summary

Computational Scientist with a PhD in Physics and over ten years of experience building and running high-performance numerical algorithms and physics-based simulations. Expertise in scientific machine learning and geospatial data analysis. Strong mathematical foundations with a track record of architecting open-source simulation libraries, leading funded research projects, and translating complex physical models into production-quality scientific software. Experienced working across cross-functional research and engineering teams.

## Skills

**Languages:** C/C++, Python, Fortran 95, Rust, R, Java, SQL, Julia, Shell
**ML & Data:** Scientific machine learning, PyTorch, TensorFlow, scikit-learn, NumPy, SciPy, Pandas, Astropy, ML-assisted parameter estimation, gradient-based optimization, SVM, random forest, neural networks
**HPC & Parallel:** OpenMP, OpenACC (GPU offloading), MPI, SLURM, PBS, HDF5
**Tools:** Git, Docker, Azure, Jenkins, Kafka, Jira, QGIS, ParaView, VisIt

## Experience

**Independent Contractor**, *Mercor* Jan 2026 – Present
Physics subject-matter expert evaluating frontier language models for a leading AI lab.

- Reviewed 50+ expert-level problems in physics, hydrodynamics, astrophysics, and cosmology; wrote original prompts that stumped frontier models.

**Spatial Data Scientist**, *TealWaters* May 2025 – Jan 2026
Built geospatial ML pipelines and optimized terrain-analysis code for wetland mapping.

- Optimized the terrain-modeling code that computes elevation derivatives from 1 m, 3 m, and 10 m DEMs over 20 tiles, improving its computational performance.
- Mapped wetland probability across the 2,160 km² Skykomish watershed with the Wetland Intrinsic Potential (WIP) random-forest model, reaching >75% accuracy.

**Independent Researcher & Open-Source Developer**, *Self-directed* Jan 2024 – May 2025
Continued simulation research and open-source development between industry roles.

- Co-developed Tleco, a Rust/Python toolkit that solves the Fokker–Planck equation coupled to radiative transfer for relativistic outflows (ApJ 2024).
- Developing WindsOfChange with the UW Forest Mycobiome Lab: a C++/R library for wind-driven spore dispersal over hilly forest terrain (RcppParallel, OpenMP).

**Software Engineer**, *Paychex* Apr 2022 – Jan 2024
Built and tested data-integration services between enterprise systems and Oracle E-Business Suite.

- Developed and tested production data-pipeline software (Java, Python, SQL) with CI/CD.

**Postdoctoral Research Associate**, *Rochester Institute of Technology* Feb 2021 – Apr 2022
HPC simulations of supermassive black hole binaries in an NSF-funded project.

- Led a 7-person team in an NSF-sponsored upgrade of PatchworkMHD, a C code for MHD simulations of supermassive black hole binaries, adding black hole spin with under 5% runtime overhead.
- Ran and benchmarked large production simulations on Frontera (TACC), about 10,000 node-hours, and identified performance bottlenecks in the parallel code.
- Calibrated Paramo parameters against Fermi-LAT observations using gradient-based optimization, comparing simulation output with data and updating the model automatically; supervised a graduate student on the same problem.
- Collaborated on a multi-institutional HPC project on binary neutron star mergers, resulting in 2 publications.

**Postdoctoral Research Fellow**, *Purdue University* Oct 2018 – Nov 2020
Radiative-transfer modeling of relativistic jets (blazars and gamma-ray burst afterglows).

- Designed and wrote Paramo, an open-source Fortran 95 library that solves the Fokker–Planck equation for relativistic particle distributions coupled to radiative transfer. Optimized with OpenMP; GPU offload tested with OpenACC.
- Implemented implicit PDE solvers, numerical integration, and arbitrary-shape particle distributions without slowing the simulation.
- Wrote Python tools to analyze Paramo output and run parameter optimization; other groups have adopted Paramo, including Combi & Siegel (2023).

- Co-I on two NASA Fermi Guest Investigator proposals, one funded ($68K) to model blazar radiation; mentored 3 graduate students and published 2 peer-reviewed papers.

**Postdoctoral Research Fellow**, *Universidad Michoacana de San Nicolás de Hidalgo* Jan 2018 – Sep 2018

- Developed a Python pipeline to process black hole simulation outputs from the GRTrans numerical code, generating SVM training data for radio image prediction.
- Built open-source data analysis and visualization tools in Python for radiative transfer phenomena in relativistic astrophysics scenarios.

**Graduate Research Assistant**, *Universitat de València* Oct 2011 – Jul 2017

- Studied how magnetization of shocked plasma shapes the radiation spectra of relativistic internal shocks, using the C-SPEV simulation code.

## Projects

**Blazar Population ML** Jan 2026 – Present

*Scientific ML (Python, scikit-learn, PyTorch)* github.com/altjerue/ilwikak-ml

- Applying classical ML and Bayesian inference to the Fermi-LAT 4LAC-DR3 catalog (3,407 sources) to recover physical jet parameters from multi-wavelength data.
- Building a neural-network emulator of the physics-based Tleco model that maps physical jet parameters to observable spectral energy distributions, to be validated against Tleco output (in progress).
- Forecasting blazar gamma-ray flares with a time-series foundation model fine-tuned on Tleco-generated synthetic light curves.

## Selected Publications (9 peer-reviewed articles, 3 first-author)

- Davis, Z., **Rueda-Becerril, J. M.**, & Giannios, D. *Tleco: A Toolkit for Modeling Radiative Signatures from Relativistic Outflows*, ApJ 976, 182 (2024), arXiv:2405.17581.
- **Rueda-Becerril, J. M.**, Harrison, A. O., & Giannios, D. *Blazar jets launched with similar energy per baryon, independently of their power*, MNRAS 501, 4092–4102 (2021), arXiv:2009.02273.
- **Rueda-Becerril, J. M.**, Mimica, P., & Aloy, M. A. *On the influence of a hybrid thermal–non-thermal distribution in the internal shocks model for blazars*, MNRAS 468, 1169–1182 (2017), arXiv:1612.06383.

## Education

**Universitat de València**, *Ph.D. in Physics, Excellent Cum Laude* 2017

**Universidad Michoacana de San Nicolás de Hidalgo**, *M.Sc. in Physics*

**Universidad Autónoma del Estado de México**, *B.Sc. in Physics*